# Data structures cheat sheet

Python Lab cheat sheet. Snapshot 17 October 2026. This is a study aid, not a specification. Confirm details in the Sources section of Python Lab and on the official pages linked there.

## Sequences

- list: ordered, mutable. append is amortized constant time in CPython. Inserting at the front is linear.
- tuple: ordered, immutable, hashable when every item is hashable.
- str: immutable sequence of Unicode code points. Slicing copies.
- deque from collections: append and pop at either end are cheap. Indexing in the middle is not.
- A slice is a new object for list, tuple, and str. The source is not mutated by the slice itself.

## Maps and sets

- dict preserves insertion order. Keys must be hashable.
- Average-case lookup, insert, and delete for dict and set are constant time in CPython. Worst cases exist.
- set holds unique hashable items and is unordered for algorithm purposes.
- Use a tuple or frozenset when you need a hashable collection.

## Study patterns

- Stack: append and pop on a list.
- Queue: collections.deque, popleft.
- Binary search needs a sorted random-access sequence. bisect is in the standard library.
- A hash table is the idea behind dict. A collision strategy is an implementation detail you do not control.
- State complexity as a model of growth, and name the implementation (usually CPython) when the cost is specific.